

# Luis López Fernández

📍, St. Gallen, CH

✉️ [luislofer+work@proton.me](mailto:luislofer+work@proton.me)

☎️ +41 78 727 82 85

in [luislopezfdez](#)

## 👤 ABOUT

AI Engineer & Project Lead specializing in translating complex data into production-ready solutions. Over 10+ years bridging research and industry to ship computer vision, spatial intelligence, and generative AI systems at scale.

## 📁 WORK EXPERIENCE

**AI Engineer & Project Lead,**  
2020-07 - Present

### HIGHLIGHTS

- Architected AI solutions for efficient processing of multimodal data (point clouds, imagery, GNSS).
- Managed and executed research projects, proofs of concept, and feasibility studies.
- Designed ML infrastructure for scalable training and deployment of models in production.
- Coordinated multidisciplinary teams of engineers & researchers across departments.
- Designed and implemented generative AI solutions supporting Hexagon AB processes and product portfolio.
- Spearheaded LLM and multimodal generation initiatives for enterprise integration.

**Predoctoral/Postdoctoral Researcher & Consultant,**  
2014-06 - Present

### HIGHLIGHTS

- Led research on geomatics & machine learning applications for energy optimisation.
- Contributed to software development & scientific publications on remote sensing.
- Conducted research and consulting in photogrammetry, computer vision, and artificial intelligence.
- Collaborated on funded research projects and academic-industry partnerships.
- Provided technical guidance on data fusion & 3D modelling methods.

**CV/AI Engineer & SCRUM Master,**  
2018-03 - 2020-06

### HIGHLIGHTS

- Developed real-time LiDAR processing stack for ADAS platforms of automotive OEMs.
- Created calibration & perception algorithms for RADAR, LiDAR & camera systems.
- Facilitated Agile/Scrum processes improving team velocity & delivery quality.

**CV/AI Engineer & R&D Project Lead,**  
2016-07 - 2018-02

### HIGHLIGHTS

- Managed AI and computer vision R&D projects combining GNSS, LiDAR and imagery data.
- Designed multi-sensor calibration strategies for infrastructure inspection systems.
- Coordinated technical development teams & project stakeholders.

## SKILLS

### Generative AI & Agents

Agentic & RAG architectures   LLM observability, evaluation & benchmarking   Prompt & context engineering

LLM fine-tuning & preference alignment

### Perception and Spatial Intelligence

Image & point cloud processing   Geospatial analytics   3D computer vision and AI   Sensor fusion and calibration

### Cloud/Edge Engineering

Cloud architectures (AWS/Azure/GCP)   Edge & real-time systems   AI production pipelines

Containerization & orchestration (Docker, Kubernetes)

### Leadership & Strategy

Research-to-product execution   Portfolio & project leadership   Cross-functional team coordination

Stakeholder management

## EDUCATION

**PhD: Geotechnologies applied to Construction, Energy and Industry, Geospatial Technologies - University of Salamanca**  
2013-Sep - 2017-Jul

**MSc: Cartographic Geotechnologies applied to Engineering and Architecture, Geospatial Technologies - University of Salamanca**  
2012-Sep - 2013-Jul

**BSc: Surveying Engineering, Geospatial Technologies - University of Oviedo**  
2007-Sep - 2011-Jul

**VET: Advanced Vocational Training in Multiplatform & Web Application Development, Software Development - C.I.F.P. de Avilés**  
2013-Aug - 2016-Jun

## PUBLICATIONS

**Photomatch** — Open-source multi-view and multi-modal feature matching tool for photogrammetric applications.

**GRAPHOS** — Open-source software for photogrammetric applications.

**Feature matching evaluation for multimodal correspondence.**

**Automatic evaluation of photovoltaic power stations from high-density RGB-T 3D point clouds.**

**Large-scale automatic analysis and classification of roof surfaces for solar panel installation using multi-sensor aerial platforms.**

**Thermographic and mobile indoor mapping for energy-loss computation in buildings.**

**See full publication list online**  
**Present**

Full publication list online.